

AWS

COMPLETE NOTES

BEGINNER → ADVANCED



All Important AWS Services
in One Place



Compute
EC2, Lambda



Storage
S3, EBS



Database
RDS, DynamoDB



Security
IAM, KMS



Monitoring
CloudWatch

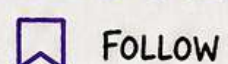
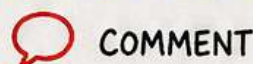
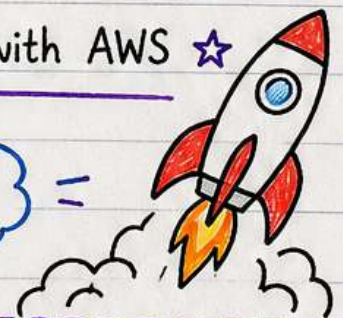


Networking
VPC, Route 53

EC2 • S3 • RDS • VPC • IAM • LAMBDA • CLOUDWATCH • ECS • EKS

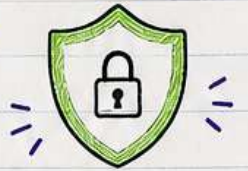
★ Your Complete Guide to Cloud Computing with AWS ★

"Learn → Practice → Build → Grow"

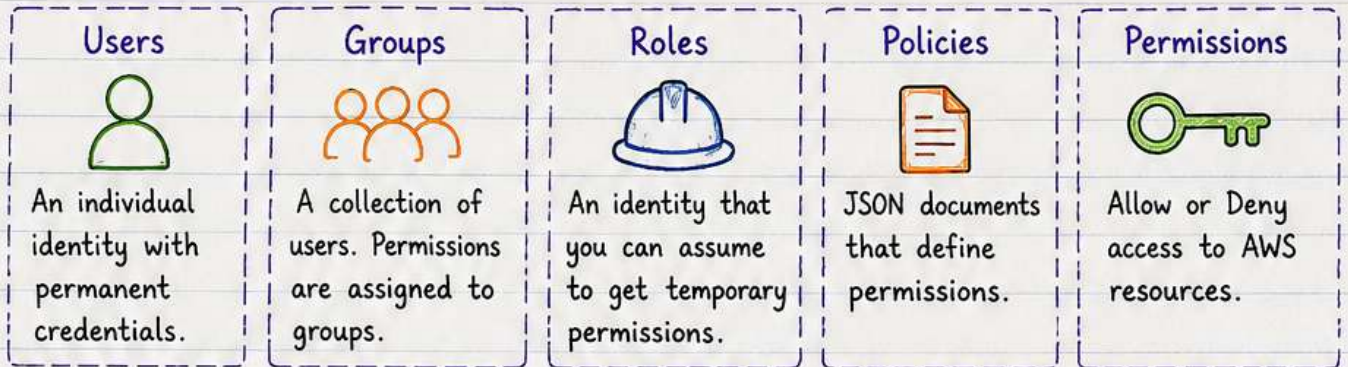


2. AWS IAM (Identity and Access Management)

AWS IAM helps you securely control access to AWS services and resources.

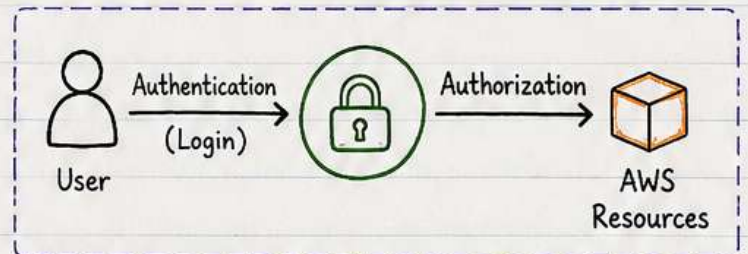


★ 1. KEY CONCEPTS



2. AUTHENTICATION vs AUTHORIZATION

- Authentication → Who are you? (Proving your identity)
- Authorization → What can you do? (Access permissions)



3. ROOT USER

- Created when you create AWS account.
- Has full access to all AWS services.
- Best practice → Do not use root user for daily activities.
- Enable MFA for root user.

4. MFA (Multi-Factor Authentication)

- Adds extra layer of security.
- Requires 2 or more verification methods.

Types → Virtual MFA (Authenticator App)
→ Hardware MFA (YubiKey)
→ SMS MFA



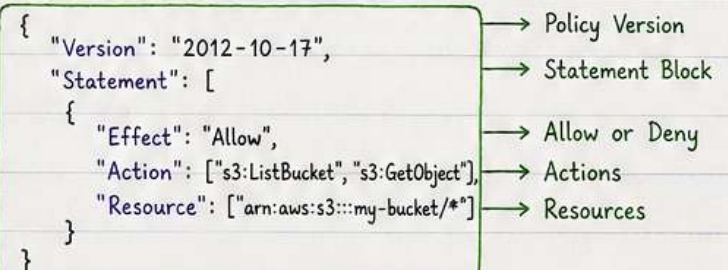
5. LEAST PRIVILEGE PRINCIPLE

"Grant only the minimum permissions necessary to perform the job."

→ Reduces risk and improves security.



6. IAM POLICY STRUCTURE



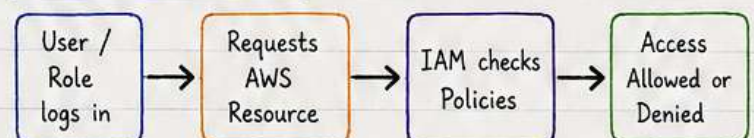
7. POLICY EFFECT

Allow → Grants permission

Deny → Explicitly blocks access

★ Deny always overrides Allow.

8. HOW IAM WORKS



LIKE



SHARE



COMMENT



SAVE

@abhi_techhub

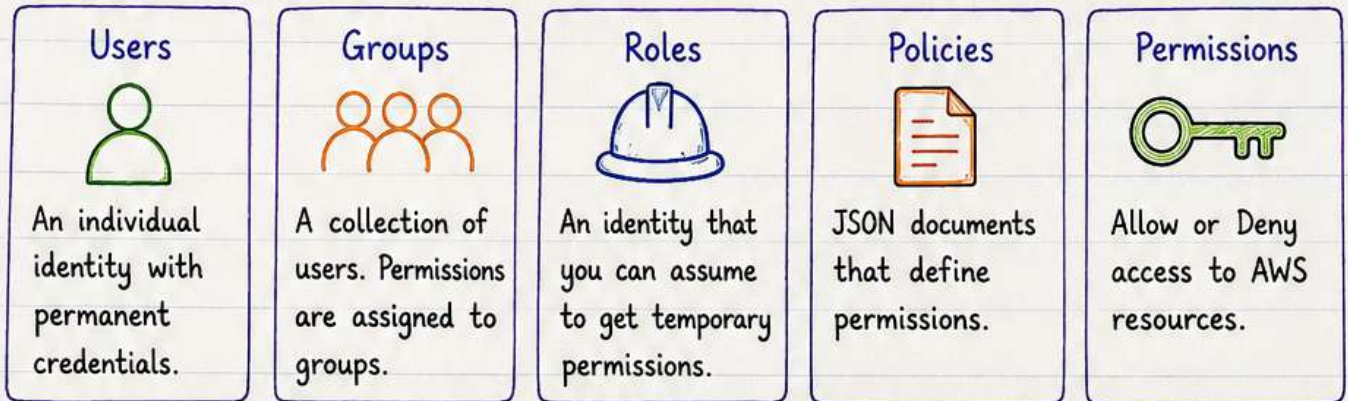


2. AWS IAM (Identity and Access Management)

AWS IAM helps you securely control access to AWS services and resources.

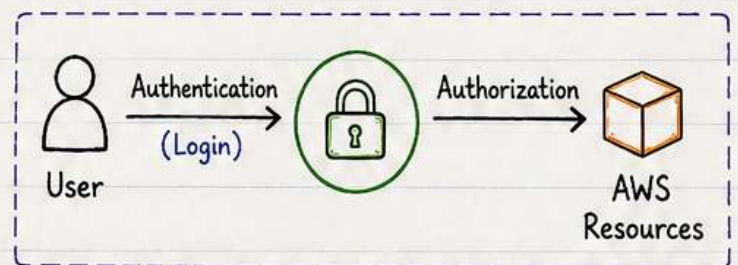


★ 1. KEY CONCEPTS



2. AUTHENTICATION vs AUTHORIZATION

- Authentication → Who are you?
(Proving your identity)
- Authorization → What can you do?
(Access permissions)



3. ROOT USER

- Created when you create AWS account.
- Has full access to all AWS services.
- Best practice → **Do not use root user for daily activities.**
- Enable MFA for root user.



4. MFA (Multi-Factor Authentication)

- Adds extra layer of security.
- Requires 2 or more verification methods.
- Types → 1. Virtual MFA (Authenticator App)
2. Hardware MFA (YubiKey)
3. SMS MFA



5. LEAST PRIVILEGE PRINCIPLE

"Grant only the minimum permissions necessary to perform the job."

→ Reduces risk and improves security.



6. IAM POLICY STRUCTURE

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["s3:ListBucket", "s3:GetObject"],
      "Resource": ["arn:aws:s3:::my-bucket/*"]
    }
  ]
}
  
```

→ Policy Version
→ Statement Block
→ Allow or Deny
→ Actions
→ Resources

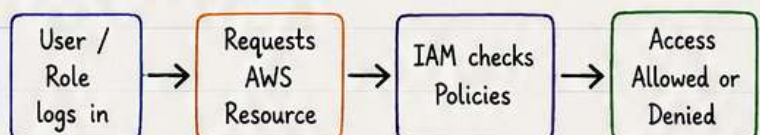
7. POLICY EFFECT

Allow → Grants permission

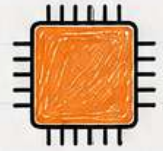
Deny → Explicitly blocks access

★ Deny always overrides Allow.

8. HOW IAM WORKS

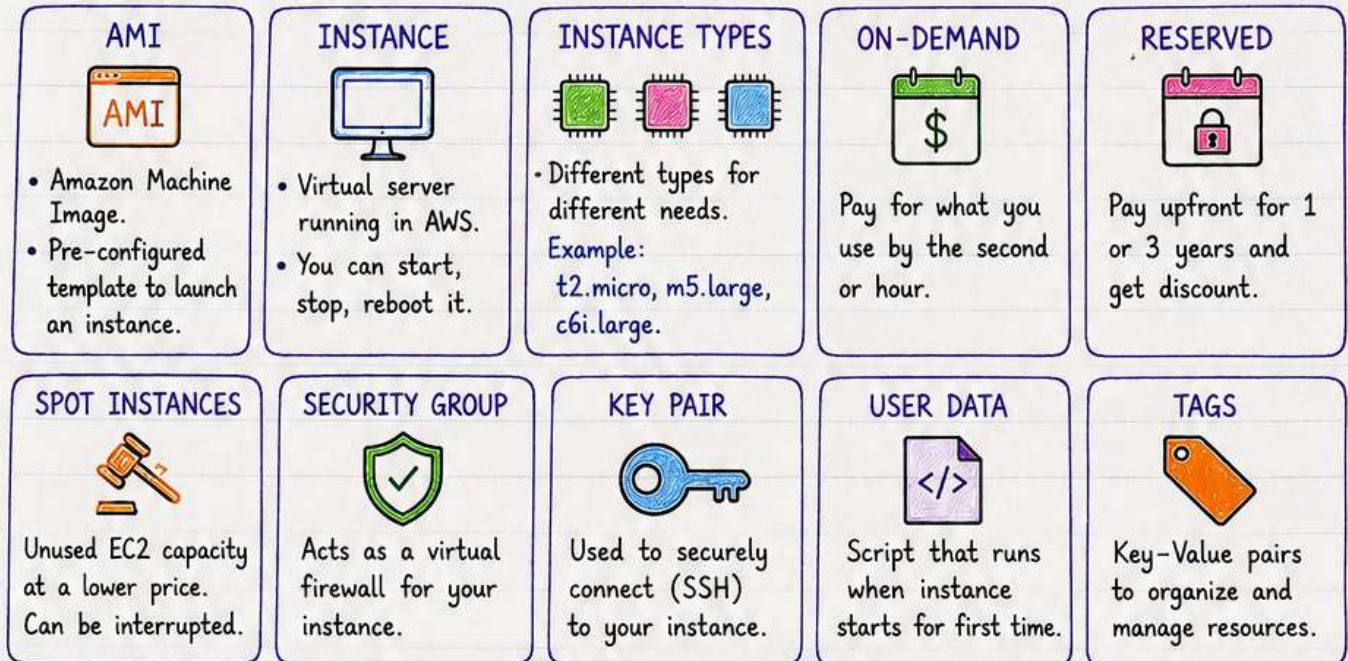


3. EC2 - Compute (Elastic Compute Cloud)

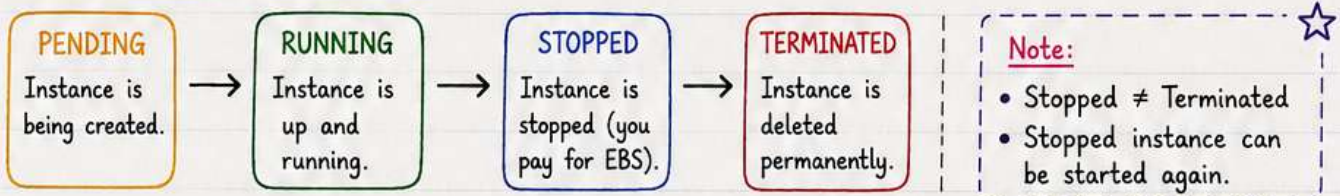


Amazon EC2 provides resizable compute capacity in the cloud. You can launch virtual servers (instances) and run applications.

★ 1. KEY CONCEPTS



2. EC2 INSTANCE LIFECYCLE



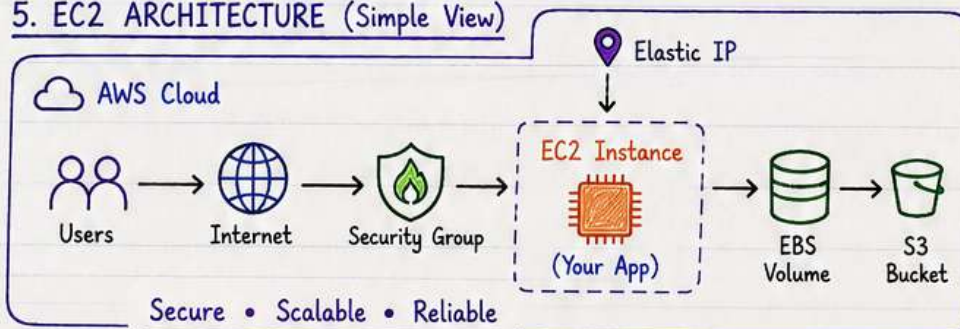
3. PRICING OPTIONS

- On-Demand - Flexible, Pay as you go.
- Reserved Instances - 1 or 3 years commitment.
- Spot Instances - Up to 90% cheaper.
- Dedicated Hosts - Physical server dedicated to you.
- Saving Plans - Commit usage (1 or 3 years).
- Capacity Reservations - Reserve capacity in advance.

4. EC2 STORAGE OPTIONS

- EBS** (Elastic Block Store) → Persistent block storage attached to instance.
- Instance Store → Temporary storage (lost when instance stop/terminate).
- EFS** (Elastic File System) → Shared file storage for Linux instances.
- S3** - Object storage (not mounted as block).

5. EC2 ARCHITECTURE (Simple View)



Use Cases

- Web Servers
- Application Servers
- Backend Services
- Batch Processing
- Dev / Test Environments

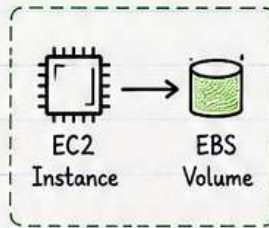
4. EC2 - ADVANCED

Advanced features and services that work with EC2 to build highly available and scalable applications.



1. EBS (ELASTIC BLOCK STORE)

- Persistent block storage for EC2.
- Data is safe even if instance stops.
- Types: gp3 (General Purpose), io1/io2 (Provisioned IOPS), st1 (HDD), sc1 (Cold HDD).
- Snapshots → Backup of EBS.



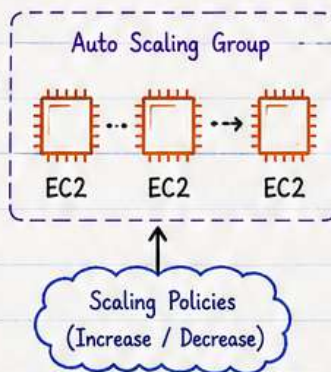
2. ELASTIC IP

- Static public IPv4 address.
- Remains same even if instance restarts.
- Can be associated and disassociated.



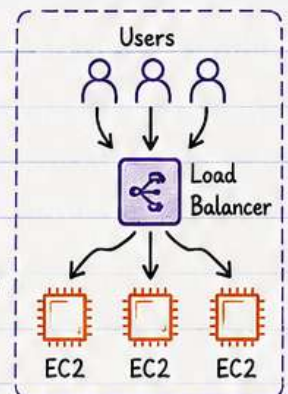
3. AUTO SCALING

- Automatically adjusts capacity of EC2 instances.
- Based on conditions (CPU, Network, Custom metrics).
- Helps in cost optimization and high availability.



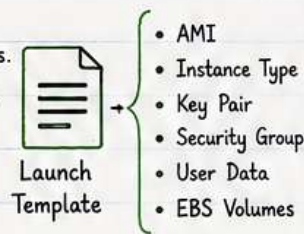
4. LOAD BALANCING

- Distributes traffic across multiple instances.
- Improves availability and reliability.
- Types:
 - ALB (Application Load Balancer)
 - NLB (Network Load Balancer)
 - CLB (Classic Load Balancer)



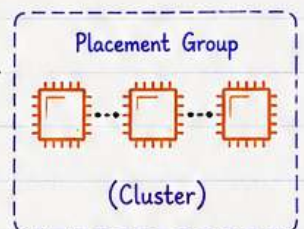
5. LAUNCH TEMPLATES

- Blueprints for launching EC2 instances.
- Includes AMI, instance type, key pair, security group, user data, etc.
- Used by Auto Scaling and EC2 Fleet.



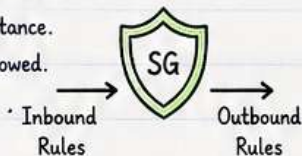
6. PLACEMENT GROUPS

- Control how instances are placed on underlying hardware.
- Types:
 - Cluster → Low latency
 - Spread → High availability
 - Partition → Large distributed applications (HDFS, Kafka)



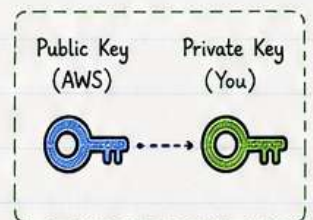
7. SECURITY GROUPS (RECAP)

- Acts as virtual firewall for instance.
- Stateful - Return traffic is allowed.
- Rules: Allow / Deny traffic.
- Applied at instance level.



8. KEY PAIR

- Used for SSH access to EC2.
- Public key → Stored by AWS.
- Private key (.pem) → You download and keep secure.



9. LOAD BALANCER COMPARISON

Feature	ALB (Application)	NLB (Network)	CLB (Classic)
Layer	Layer 7 (HTTP/HTTPS)	Layer 4 (TCP/UDP)	Layer 4 & 7
Protocol	HTTP, HTTPS, WebSocket	TCP, UDP, TLS	HTTP, HTTPS, TCP
Performance	Good	Very High	Low (Legacy)
Use Case	Web Apps, Microservices	High Performance Apps	Legacy Applications
Features	Path-based routing, Host routing	Static IP, Ultra low latency	Sticky sessions

IMPORTANT TIP

Use the right combination of EC2 features to build secure, scalable, and cost-effective applications on AWS!



LIKE



SHARE



COMMENT



SAVE

@abhi_techhub



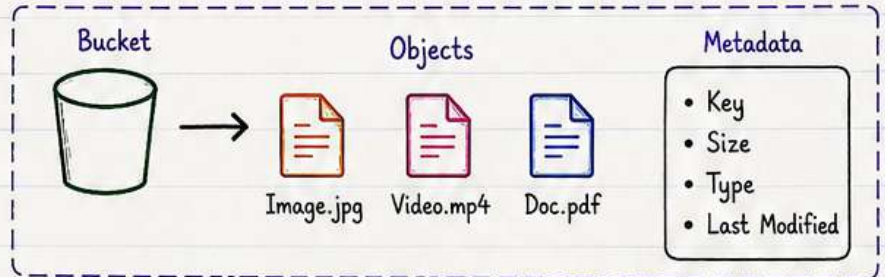
5. S3 - OBJECT STORAGE

Amazon S3 (Simple Storage Service) is an object storage service that offers industry-leading scalability, data availability, security and performance.



1. WHAT IS S3?

- Stores data as objects inside buckets.
- Each object consists of data, metadata and a unique key.
- Access through AWS Management Console, CLI or SDK.



2. STORAGE CLASSES

S3 STANDARD

- General purpose
- Frequent access
- Low latency
- High durability

S3 INTELLIGENT-TIERING

- Automatic cost optimization
- Moves data between tiers automatically

S3 STANDARD-IA (INFREQUENT ACCESS)

- Lower cost
- For infrequent access
- Retrieval charges

S3 ONE ZONE-IA

- Lower cost
- Data in single Availability Zone
- Recreated if AZ is destroyed

S3 GLACIER

- Very low cost
- Archival storage
- Retrieval takes minutes to hours

S3 GLACIER DEEP ARCHIVE

- Lowest cost
- Retrieval takes hours
- Long term archive

3. BUCKET FEATURES

- Global namespace → Bucket names must be unique worldwide.
- Versioning → Keep multiple versions of an object.
- Lifecycle Rules → Automatically transition or delete objects.
- Encryption → Protect data at rest using SSE-S3, SSE-KMS, or CSE-KMS.
- Bucket Policies → Control access to buckets and objects.
- CORS → Allow cross-origin access.

4. S3 DATA CONSISTENCY

- Strong read-after-write consistency for PUTS and DELETES.
- List operations are also strongly consistent.
- Ensures the latest data is always returned.

5. ACCESS METHODS

- AWS Management Console
- AWS CLI
- AWS SDK (Language APIs)
- REST API

6. SECURITY

- IAM Policies → Control who can access S3.
- Bucket Policies → Set permissions at bucket level.
- Encryption → Protect data in transit and at rest.
- MFA Delete → Extra protection for versioned buckets.
- Block Public Access → Prevent public exposure.

7. S3 KEY FEATURES



Highly Scalable
Store unlimited amount of data.



Highly Durable
99.999999999% (11 nines) durability.



Highly Available
Designed for 99.99% availability.



Cost Effective
Pay for what you use.



Performance
High throughput and low latency.

8. COMMON USE CASES

- Backup and Restore
- Data Archiving
- Static Website Hosting
- Big Data Analytics
- Media Hosting
- Disaster Recovery



S3 is the foundation of many AWS services and a core building block for secure, scalable and cost-effective cloud storage.



LIKE



SHARE



COMMENT



SAVE

@abhi_techhub



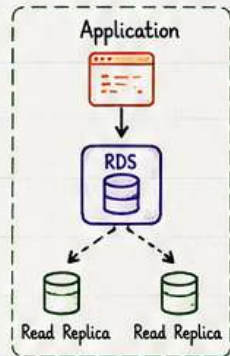
6. AWS DATABASE SERVICES

AWS provides a wide range of database services to store, manage and scale your data in the cloud.



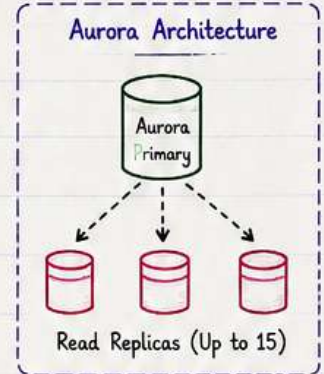
1. RELATIONAL DATABASE SERVICE (RDS)

- Managed relational database service.
- Supports: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server.
- Automated backups, patching, monitoring.
- Multi-AZ for High Availability.
- Read Replicas for scaling reads.



2. AMAZON AURORA

- MySQL and PostgreSQL compatible.
- High performance and availability.
- Auto-scaling storage.
- Up to 15 read replicas.
- Multi-AZ DB cluster.



3. DYNAMODB (NoSQL)

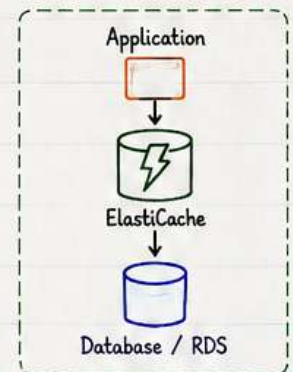
- Fully managed NoSQL database.
- Key-Value and Document database.
- Single digit millisecond latency.
- Automatically scales.
- Uses capacity units (RCU / WCU).
- Durable and highly available.

DynamoDB Table		
Partition Key	Sort Key	Attributes
UserID = 101	OrderID = 1	Name, Price
UserID = 101	OrderID = 2	Name, Price
UserID = 102	OrderID = 1	Name, Price
...	...	...

Partition Key + Sort Key = Unique Item

4. ELASTICACHE

- In-memory data store.
- Improves application performance.
- Supports:
 - Redis
 - Memcached
- Used for caching, session store, leaderboards etc.

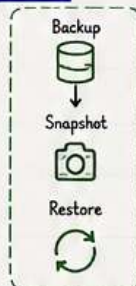


5. DATABASE COMPARISON

Service	Type	Use Case	Scaling	Best For
RDS	Relational	Traditional applications	Vertical + Read Replicas	Web Apps, ERP, CRM
Aurora	Relational (Cloud Native)	High performance apps	Auto Scaling + Replicas	Modern Apps, SaaS
DynamoDB	NoSQL (Key-Value / Document)	Large scale, low latency	Automatic	Mobile, IoT, Gaming
ElastiCache	In-Memory Cache	Caching, session store	Automatic	High speed apps

6. BACKUPS & RECOVERY (RDS & AURORA)

- Automated Backups → Enabled by default.
- Manual Snapshots → On-demand backups.
- Point-in-Time Recovery (PITR) → Restore to any second.
- Cross-Region Snapshot → Disaster Recovery.



7. DYNAMODB KEY CONCEPTS

- Partition Key → Determines partition.
- Sort Key → Sorts data within the same partition.
- Indexes:
 - GSI (Global Secondary Index)
 - LSI (Local Secondary Index)

Example	
PK (UserID)	SK (OrderID)
101	1
101	2
102	1

All items with same PK are stored together.

8. WHEN TO USE WHAT?



- Need ACID transactions? → Use RDS / Aurora
- Need extreme performance & scale? → Use Aurora
- Need key-value / document store? → Use DynamoDB
- Need caching for faster access? → Use ElastiCache

9. DATABASE BEST PRACTICES

- ✓ Use appropriate DB for the right use case.
- ✓ Enable Multi-AZ for production workloads.
- ✓ Regular backups and test restore.
- ✓ Monitor with CloudWatch.
- ✓ Encrypt data at rest and in transit.



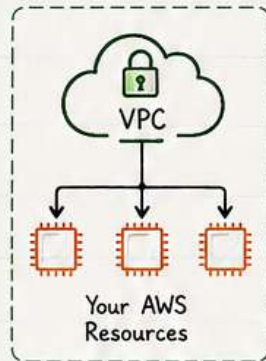
7. AWS VPC & NETWORKING

Amazon VPC (Virtual Private Cloud) lets you launch AWS resources in a logically isolated network that you define.



1. WHAT IS VPC?

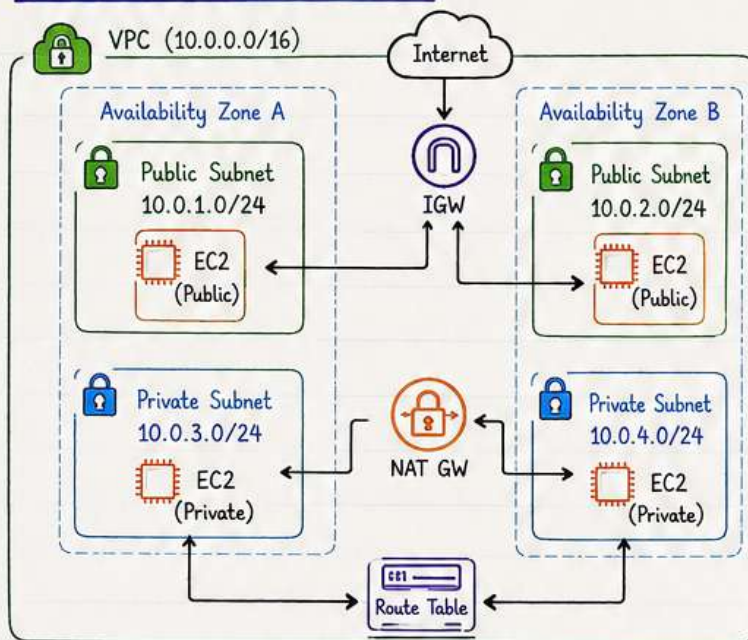
- A virtual network dedicated to your AWS account.
- You have complete control over IP range, subnets, route tables and gateways.
- Isolated from other VPCs.
- Highly secure and scalable.



2. VPC COMPONENTS

	VPC	Your virtual network in AWS.
	Subnet	A range of IP addresses in a VPC.
	Route Table	Contains rules that determine where network traffic is routed.
	Internet Gateway	Enables communication between VPC and internet.
	NAT Gateway	Allows private subnet instances to access internet.
	Security Group	Stateful firewall for instances.
	Network ACL	Stateless firewall for subnets.

3. VPC ARCHITECTURE (EXAMPLE)



4. SUBNETS

- Public Subnet → Has route to Internet Gateway. Resources can be accessed from internet.
- Private Subnet → No direct route to Internet. Access via NAT Gateway.

Type	Use Case
Public Subnet	Web Servers, Bastion Host, Load Balancer
Private Subnet	Application Servers, Databases, Internal Services

5. ROUTE TABLE EXAMPLE

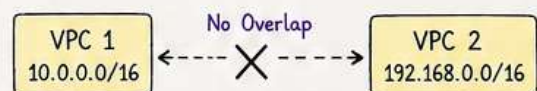
Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	Internet Gateway (Public Subnet)
0.0.0.0/0	NAT Gateway (Private Subnet)

6. SECURITY GROUP vs NETWORK ACL

	Security Group (SG)	Network ACL (NACL)
Type	Virtual Firewall for EC2	Virtual Firewall for Subnet
Level	Instance Level	Subnet Level
Rule Type	Allow Rules Only	Allow & Deny Rules
State	Stateful	Stateless
Default	Allow All Outbound	Deny All
Applies To	ENI / Instance	Subnet
Order	Not Applicable	Rule Number (Top to Bottom)

7. CIDR BLOCKS

- CIDR (Classless Inter-Domain Routing) defines the IP range of your VPC.
- Example: 10.0.0.0/16 → 65,536 IPs
- Choose non-overlapping CIDR blocks across VPCs.



8. BEST PRACTICES

- ✓ Use Private Subnets for databases and internal resources.
- ✓ Use Security Groups to allow only necessary traffic.
- ✓ Use NACLs for an additional layer of security.
- ✓ Use NAT Gateway for private subnet internet access.
- ✓ Enable VPC Flow Logs for monitoring traffic.
- ✓ Use appropriate CIDR blocks and plan IP addressing.



9. COMMON USE CASES

- Host web applications in public subnets.
- Keep databases in private subnets.
- Connect on-premise network using VPN / Direct Connect.
- Build multi-tier architectures.



LIKE



SHARE



COMMENT



SAVE

@abhi_techhub



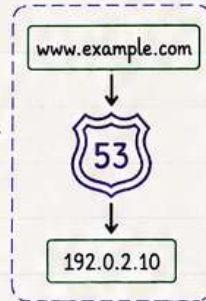
8. AWS NETWORKING - ADVANCED

AWS provides powerful networking services to connect your resources, users and applications securely around the world.



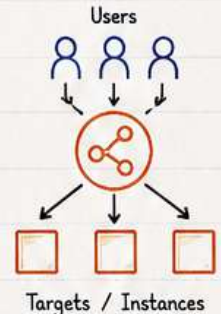
1. ROUTE 53 (DNS SERVICE)

- Scalable Domain Name System (DNS) web service.
- Converts domain names to IP addresses.
- Highly available and low latency.
- Health checks, traffic routing, domain registration.



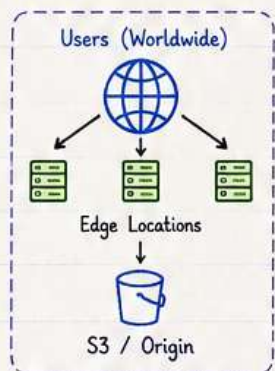
2. ELASTIC LOAD BALANCING

- Automatically distributes incoming traffic across multiple targets.
- Improves application availability, fault tolerance and scalability.
- Types: ALB, NLB, GWLB, CLB



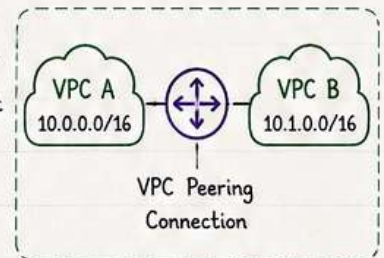
3. CLOUD FRONT (CDN)

- Content Delivery Network.
- Delivers content with low latency and high transfer speeds.
- Global Edge Locations cache your content.
- DDoS protection, SSL/TLS, compression, geo-restriction.



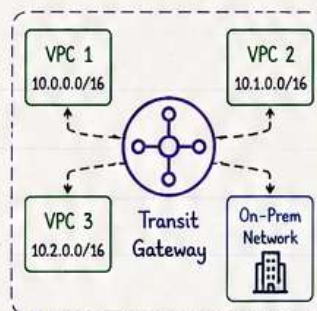
4. VPC PEERING

- Connects two VPCs privately using AWS backbone network.
- No transit gateway or internet gateway required.
- Supports overlapping CIDR (with caution).



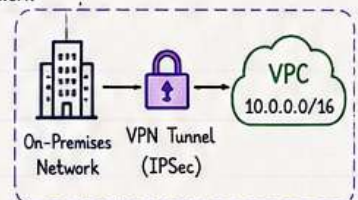
5. TRANSIT GATEWAY

- Connects multiple VPCs and on-premises networks using a central hub.
- Scales to thousands of VPCs.
- Simplifies network management.
- Supports inter-region peering.



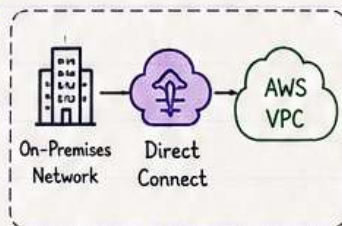
6. VPN CONNECTION

- Connect your on-premises network to AWS over the internet.
- Encrypted tunnel using IPSec.
- Types:
 - Site-to-Site VPN
 - Client VPN



7. AWS DIRECT CONNECT

- Dedicated network connection from on-premises to AWS.
- More consistent network experience than internet.
- Reduced bandwidth costs.
- Higher throughput and security.

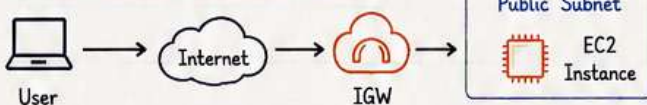


8. ROUTE 53 ROUTING POLICIES

- Simple** → Route traffic to a single resource.
- Weighted** → Distribute traffic based on weights.
- Latency** → Route to lowest latency region.
- Failover** → Active-passive setup.
- Geolocation** → Route based on user location.
- Geoproximity** → Route based on resources near user.

9. PUBLIC vs PRIVATE TRAFFIC FLOW

A. Public Subnet (via Internet Gateway)



B. Private Subnet (via NAT Gateway)



10. NETWORK ACL vs SECURITY GROUP

Feature	Network ACL (NACL)	Security Group (SG)
Type	Stateless	Stateful
Level	Subnet Level	Instance Level
Rule	Allow / Deny	Allow Only
Default	Deny All	Allow All Outbound
Applies To	Subnet	EC2 Instance
Order	Rule Number	Not Applicable



LIKE



SHARE



COMMENT



SAVE

@abhi_techhub



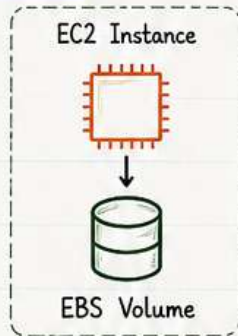
9. AWS STORAGE & FILE SERVICES

AWS provides a variety of storage and file services to store, share and manage data securely and cost-effectively.



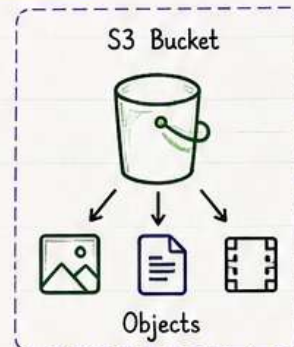
1. EBS (ELASTIC BLOCK STORE)

- Block level storage for EC2.
- Persists independently of instance.
- High performance.
- Types: gp3, io1/io2, st1, sc1.
- Use case: Databases, critical applications.



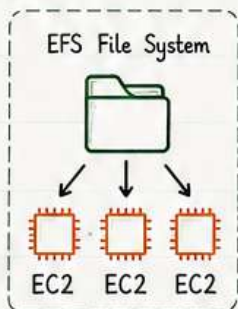
2. S3 (SIMPLE STORAGE SERVICE)

- Object storage service.
- Highly durable (11 9s durability).
- Scalable and cost-effective.
- Stores any amount of data.
- Use cases: Backup, static website, data lake, archives.



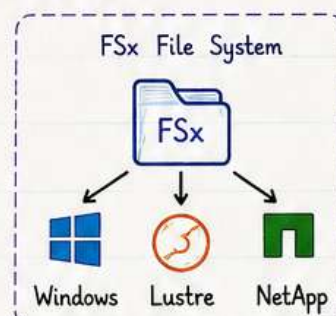
3. EFS (ELASTIC FILE SYSTEM)

- Managed file storage for Linux instances.
- Shared across multiple EC2 instances.
- Uses NFS (v4.1 & v4.2) protocol.
- Use case: Content management, web serving.



4. FSx (FULLY MANAGED FILE SYSTEMS)

- Managed file systems for Windows, Lustre, NetApp.
- High performance and scalability.
- Integrated with AWS services.
- Use case: Windows shares, HPC, enterprise applications.



5. S3 STORAGE CLASSES

Storage Class	Use Case	Durability	Availability
S3 Standard	Frequently accessed data	11 9s	99.99%
S3 Standard-IA	Infrequently accessed	11 9s	99.9%
S3 One Zone-IA	Recreatable data	11 9s	99.5%
S3 Glacier Instant	Archive, instant retrieval	11 9s	99.9%
S3 Glacier Flexible	Archive, minutes-hours	11 9s	99.9%
S3 Glacier Deep Archive	Long term archive (lowest cost)	11 9s	99.9%

TIP

Choose the right storage class to optimize cost and performance based on access patterns.



6. EBS VOLUME TYPES

Type	Best For	Characteristics
gp3	General Purpose	Balanced performance and cost.
io1 / io2	High IOPS	High performance, mission critical apps.
st1	Throughput Intensive	Low cost, high throughput, Big data, logs.
sc1	Cold Workloads	Lowest cost, infrequently accessed data.

7. KEY TAKEAWAYS

- ✓ EBS → Block storage for EC2.
- ✓ S3 → Object storage for any type of data.
- ✓ EFS / FSx → File storage for shared access.
- ✓ Choose correct storage class / type to optimize cost.
- ✓ AWS storage is durable, secure and scalable.



8. STORAGE COMPARISON

Service	Type	Access	Shared	Protocol	Use Case
EBS	Block	Single AZ	No	N/A	Databases, OS
S3	Object	Multi AZ	Yes	HTTP/HTTPS	Backup, Archives
EFS	File	Multi AZ	Yes	NFS	Shared Files (Linux)
FSx	File	Multi AZ	Yes	NFS / SMB	Windows, HPC, Enterprise

QUICK RECAP



- Block → EBS
- Object → S3
- File (Linux) → EFS
- File (Windows/Other) → FSx



LIKE



SHARE



COMMENT



SAVE

@abhi_techhub



10. AWS SECURITY & IAM

AWS provides a shared responsibility model and a wide range of security services to protect your resources, data and applications.



1. SHARED RESPONSIBILITY MODEL

AWS (SECURITY OF THE CLOUD)	CUSTOMER (SECURITY IN THE CLOUD)
- Data centers - Hardware - Networking - Virtualization - Managed services 	- Data - Applications - OS, network config - Access management - User permissions 

2. IAM (IDENTITY & ACCESS MANAGEMENT)

- Manage access to AWS services and resources securely.
- Best Practice: Grant least privilege access.
- Use MFA for additional security.
- Never use root account for daily tasks.

IAM COMPONENTS

-  User
-  Group
-  Role
-  Policy
-  Credential

3. IAM POLICY STRUCTURE

- JSON based policy document.
- Defines permissions and restrictions.
- Main elements:
 - Effect (Allow / Deny)
 - Action (API/Service)
 - Resource (ARN)
 - Condition (Optional)

POLICY EXAMPLE

```
{
  "Effect": "Allow",
  "Action": ["s3:ListBucket"],
  "Resource": "arn:aws:s3:::my-bucket",
  "Condition": {
    "IpAddress": {
      "aws:SourceIp": "203.0.113.0/24"
    }
  }
}
```

4. SECURITY BEST PRACTICES

- ☒ Enable MFA for all users.
- ☒ Use IAM roles instead of long-term credentials.
- ☒ Follow least privilege principle.
- ☒ Rotate access keys regularly.
- ☒ Enable CloudTrail and monitor activities.
- ☒ Encrypt data at rest and in transit.
- ☒ Use Security Groups and NACLs to control traffic.
- ☒ Keep your AWS resources updated.

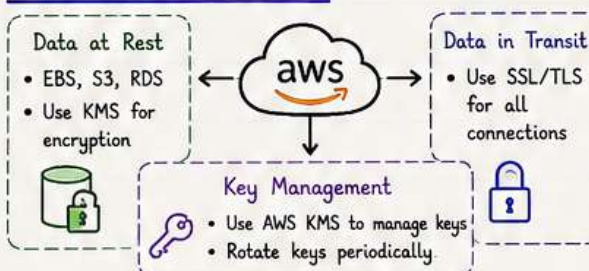
5. KEY SECURITY SERVICES

 AWS IAM	Manage users, groups, roles, policies and permissions.
 AWS CloudTrail	Records account activity and API calls for auditing.
 AWS KMS	Manage encryption keys for data protection.
 AWS WAF	Protects web applications from common exploits and attacks.
 Amazon GuardDuty	Threat detection service using ML and anomaly detection.

6. SECURITY GROUP vs NACL

Feature	Security Group (SG)	Network ACL (NACL)
Type	Stateful	Stateless
Level	Instance Level	Subnet Level
Rule	Allow Rules Only	Allow & Deny Rules
Default	Deny All	Allow All
Applies To	EC2 Instance	Subnet
Order	Not Applicable	Rule Number (1-32766)
Return Traffic	Automatically Allowed	Must be Allowed Explicitly

7. ENCRYPTION OPTIONS



8. COMMON MISTAKES

- ✗ Using root account for daily operations.
- ✗ Over-permissioned IAM policies.
- ✗ Hardcoding AWS access keys in code.
- ✗ Not enabling MFA.
- ✗ Not monitoring with CloudTrail / GuardDuty.
- ✗ Leaving unused resources and open ports.



Security in AWS is not just a feature, it's your responsibility.
Plan, Protect and Monitor — Always!



LIKE



SHARE



COMMENT



SAVE

@ abhi_techhub



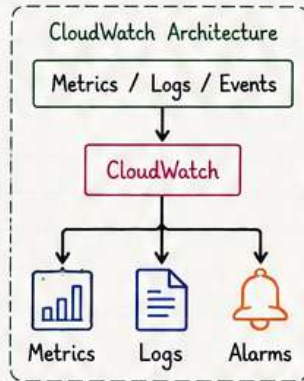
11. AWS MANAGEMENT & MONITORING

AWS provides a set of management and monitoring services to help you operate your resources efficiently.



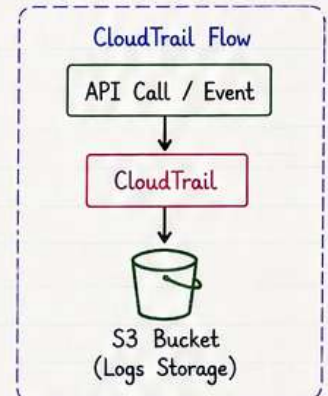
1. AMAZON CLOUDWATCH

- Monitoring and observability service.
- Collects metrics, logs and events.
- Create dashboards and alarms.
- Use cases: Performance monitoring, troubleshooting, alerting.



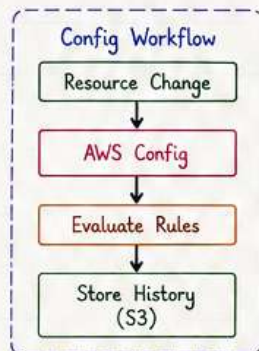
2. AWS CLOUDTRAIL

- Tracks API calls and account activity.
- Provides audit logs for governance, compliance and security.
- Logs delivered to S3 for long-term storage.



3. AWS CONFIG

- Monitors and records configuration changes of AWS resources.
- Evaluates resource configurations against desired settings.
- Helps in compliance and change management.
- Stores configuration history.



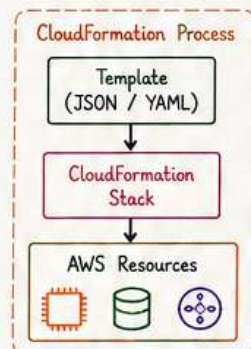
4. AWS SYSTEMS MANAGER (SSM)

- Manage and operate your EC2 instances at scale.
- Features:
 - Session Manager (SSH access without key)
 - Parameter Store
 - Patch Manager
 - Run Command
 - Automation



5. AWS CLOUDFORMATION

- Infrastructure as Code (IaC) service.
- Create and manage resources using templates (JSON / YAML).
- Ensures consistency and repeatability.
- Supports update, rollback and stack management.



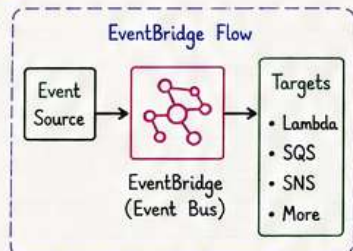
6. AWS TRUSTED ADVISOR

- Online tool to improve security, performance and cost optimization.
- Checks across 5 pillars:
 - Cost Optimization
 - Performance
 - Security
 - Fault Tolerance
 - Service Limits



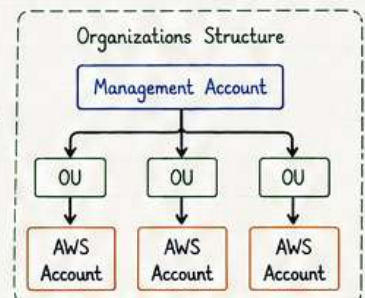
7. AMAZON EVENTBRIDGE

- Serverless event bus service.
- Connects applications using events.
- Sources: AWS services, custom apps, SaaS.
- Use cases: Event-driven architectures, automation.



8. AWS ORGANIZATIONS

- Manage multiple AWS accounts centrally.
- Create OU (Organizational Units).
- Apply policies using Service Control Policies (SCP).
- Consolidated billing and access management.



QUICK TIPS

- ✓ Use CloudWatch for monitoring & alerts.
- ✓ Enable CloudTrail for all accounts.
- ✓ Use IAM roles, not long-term credentials.
- ✓ Automate with CloudFormation & SSM.
- ✓ Review Trusted Advisor recommendations.



COMMON USE CASES

- ★ Monitor applications and infrastructure performance.
- ★ Maintain compliance and audit trails.
- ★ Automate operational tasks.
- ★ Optimize costs and resource usage.
- ★ Build secure and well-governed environments.



LIKE



SHARE



COMMENT



SAVE

@ abhi_techhub



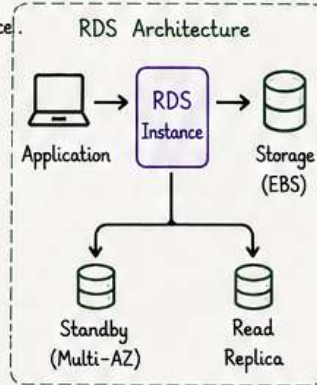
12. AWS DATABASE SERVICES

AWS offers fully managed database services for different use cases, workloads and performance needs.



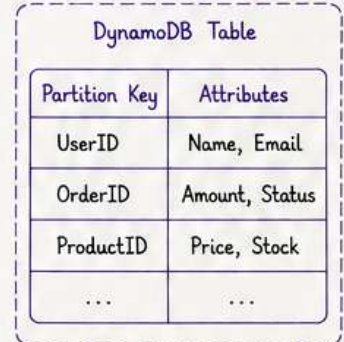
1. AMAZON RDS (RELATIONAL DATABASE SERVICE)

- Managed relational database service.
- Supports: MySQL, PostgreSQL, MariaDB, Oracle, SQL Server.
- Automated backups, patching, monitoring.
- High availability with Multi-AZ deployments.
- Use case: Web apps, ERP, traditional applications.



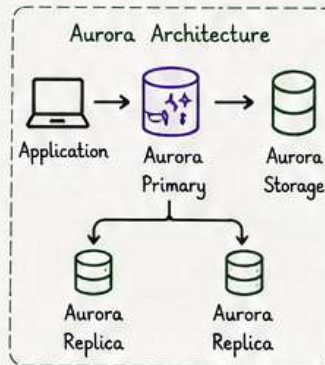
2. AMAZON DYNAMODB

- Fully managed NoSQL database.
- Key-value and document database.
- Single-digit millisecond latency.
- Auto scaling, high availability.
- Use case: Gaming, ad tech, IoT, mobile apps.



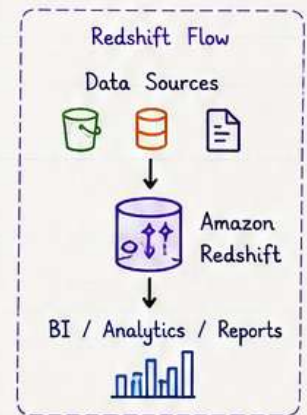
3. AMAZON AURORA

- MySQL and PostgreSQL compatible.
- Up to 5x performance of standard MySQL.
- Auto scaling storage.
- High availability with Multi-AZ.
- Use case: High performance applications.



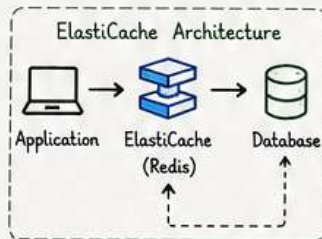
4. AMAZON REDSHIFT

- Data warehouse service for analytics.
- Columnar storage for high performance.
- Scales to petabytes of data.
- Use case: Business intelligence, reporting, data analysis.



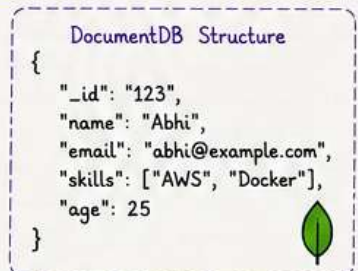
5. AMAZON ELASTICACHE

- In-memory caching service.
- Supports: Redis, Memcached.
- Improves performance and reduces database load.
- Use case: Session store, leaderboards, caching.



6. AMAZON DOCUMENTDB

- MongoDB compatible document database.
- Fully managed and highly available.
- Auto scaling storage.
- Use case: Content management, catalogs, mobile apps.



7. DATABASE COMPARISON

Service	Type	Engines	Best For	Key Features
RDS	Relational	MySQL, PostgreSQL, MariaDB, Oracle, SQL Server	General purpose applications	Multi-AZ, Read Replicas, Backups, Automated Patching
DynamoDB	NoSQL	Key-Value, Document	High scale, low latency applications	Serverless, Auto Scaling, Single-digit ms latency
Aurora	Relational	MySQL, PostgreSQL	High performance applications	Up to 5x performance, Auto Scaling, Multi-AZ
Redshift	Data Warehouse	Columnar	Analytics, Reporting, BI	Petabyte scale, Columnar Storage, High performance
ElastiCache	In-Memory Cache	Redis, Memcached	Caching, Session store, Real-time apps	Microsecond latency, High throughput, Scalable
DocumentDB	Document	MongoDB Compatible	Document based applications	Auto Scaling, High Availability, Managed

8. BEST PRACTICES

- ✓ Choose the right database for your workload.
- ✓ Enable backups and monitor performance.
- ✓ Use encryption at rest and in transit.
- ✓ Apply least privilege access.
- ✓ Review and optimize costs regularly.



 The right database can make your application faster, cheaper and more scalable!



LIKE



SHARE



COMMENT



SAVE

@ abhi_techhub



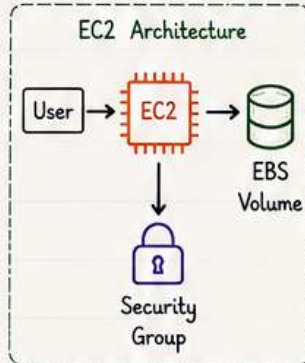
13. AWS COMPUTE SERVICES

AWS provides a wide range of compute services to run your applications, from simple VMs to serverless functions.



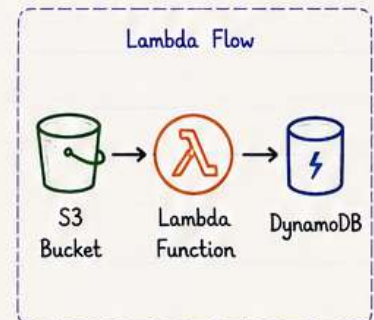
1. AMAZON EC2 (ELASTIC COMPUTE CLOUD)

- Virtual servers in the cloud.
- Full control over OS, network and storage.
- Multiple instance types and sizes.
- Use cases: Web servers, applications, dev/test, databases.



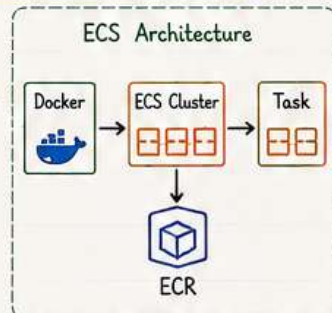
2. AWS LAMBDA

- Serverless compute service.
- Run code without managing servers.
- Pay only for the time your code runs.
- Supports multiple runtimes.
- Use cases: Microservices, file processing, automation.



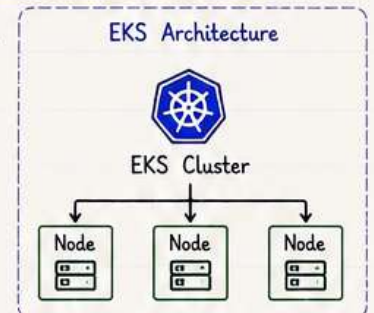
3. AWS ECS (ELASTIC CONTAINER SERVICE)

- Fully managed container orchestration service.
- Supports Docker containers.
- Easy deployment, scaling and management.
- Use case: Containerized applications.



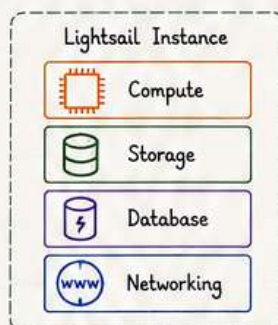
4. AWS EKS (ELASTIC KUBERNETES SERVICE)

- Managed Kubernetes service.
- Deploy, manage and scale Kubernetes clusters.
- High availability and security.
- Use case: Kubernetes workloads, microservices.



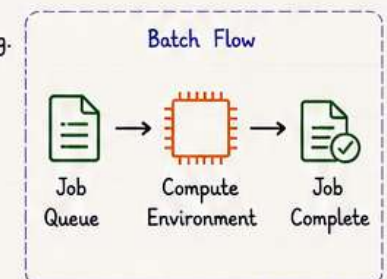
5. AWS LIGHTSAIL

- Simplified VPS (Virtual Private Server).
- Easy to launch and manage.
- Includes bundles with compute, storage, database.
- Use case: Small websites, blogs, dev environments.



6. AWS BATCH

- Fully managed batch processing.
- Runs batch computing jobs efficiently.
- Automatically provisions resources.
- Use case: Data processing, scientific simulations.



7. COMPUTE SERVICES COMPARISON

Service	Type	Management	Best For
EC2	Virtual Machines	You manage	Full control, flexibility
Lambda	Serverless	AWS manages	Event-driven, small tasks
ECS	Containers	AWS manages	Docker containers
EKS	Kubernetes	AWS manages	K8s workloads
Lightsail	VPS	AWS manages	Small websites, simple apps
Batch	Batch Processing	AWS manages	Large scale batch jobs

9. KEY TAKEAWAYS

- ✓ Choose the right compute service based on your workload.
- ✓ Serverless (Lambda) for event-driven, short tasks.
- ✓ Containers (ECS/EKS) for modern applications.
- ✓ EC2 for full control and custom configurations.
- ✓ Always consider cost, scalability and management.



8. COMMON USE CASES

- EC2** → Web servers, legacy applications, custom software.
- Lambda** → Image resizing, notifications, real-time processing.
- ECS** → Microservices, containerized applications.
- EKS** → Complex container orchestration with K8s.
- Lightsail** → Small business sites, prototypes, dev/test.
- Batch** → Data analysis, rendering, scientific workloads.

TIP

Start small, automate deployments and monitor your resources to optimize cost and performance.



LIKE



SHARE



COMMENT



SAVE

@abhi_techhub



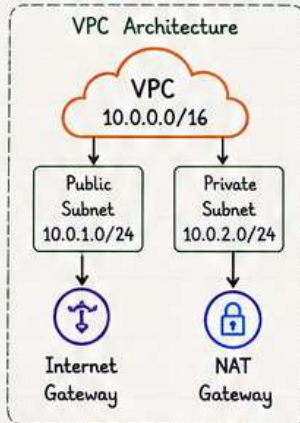
14. AWS NETWORKING SERVICES

AWS provides a variety of networking services to help you build secure, reliable and scalable applications in the cloud.



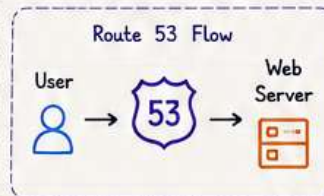
1. AMAZON VPC (VIRTUAL PRIVATE CLOUD)

- Isolated virtual network in the AWS Cloud.
- You define your IP address range.
- Complete control over networking.
- Use cases: Secure environments, hybrid cloud, microservices.



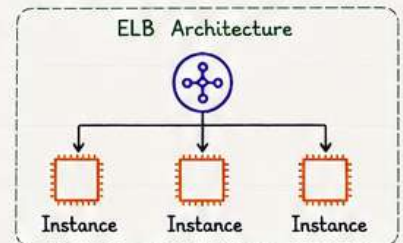
2. AMAZON ROUTE 53

- Highly available and scalable DNS service.
- Routes users to the right resources.
- Health checking & failover.
- Use cases: DNS routing, latency routing, geo routing.



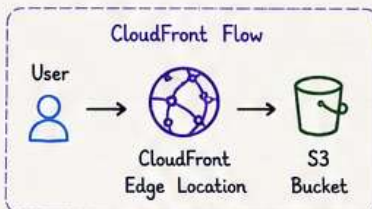
3. ELASTIC LOAD BALANCING

- Distributes incoming traffic across multiple targets.
- Improves availability and fault tolerance.
- Types: ALB, NLB, Gateway LB.
- Use cases: High traffic applications.



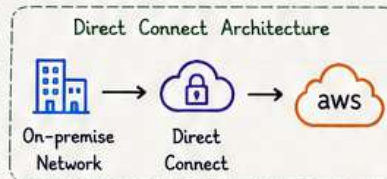
4. AMAZON CLOUD FRONT

- Content Delivery Network (CDN).
- Delivers content with low latency.
- Global edge locations.
- Use cases: Websites, streaming, APIs.



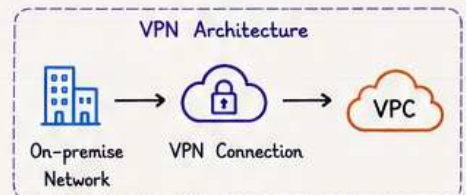
5. AWS DIRECT CONNECT

- Dedicated network connection to AWS.
- More reliable, secure and faster.
- Reduces internet traffic cost.
- Use cases: Hybrid cloud, large data transfer.



6. AWS VPN (SITE TO SITE)

- Secure encrypted connection over the internet.
- Connects your on-premise network to VPC.
- Cost-effective solution.
- Use cases: Remote offices, hybrid cloud.



7. NETWORKING SERVICES COMPARISON

Service	Type	Purpose	Best For
VPC	Networking	Isolated cloud network	All applications
Route 53	DNS	Domain name routing	Web applications
ELB	Load Balancer	Distribute incoming traffic	High availability apps
CloudFront	CDN	Deliver content globally	Websites, media
Direct Connect	Dedicated Network	Private connection to AWS	Enterprise, Hybrid cloud
VPN	Encrypted Network	Secure connection over internet	Remote offices

8. SECURITY IN NETWORKING

- ☒ Use Security Groups to control inbound/outbound traffic.
- ☒ Use NACLs for subnet level traffic filtering.
- ☒ Enable VPC Flow Logs for monitoring.
- ☒ Use AWS WAF with CloudFront and ALB.
- ☒ Encrypt data in transit (TLS) and at rest.
- ☒ Use Private Subnets for backend resources.



9. COMMON USE CASES

- Build a secure and isolated network for your applications using VPC.
- Route users to the nearest instance using Route 53.
- Handle millions of users by distributing traffic with ELB.
- Deliver static and dynamic content globally using CloudFront.
- Establish private, dedicated connection to AWS using Direct Connect.
- Connect remote offices securely using VPN.



10. KEY TAKEAWAYS

- ✓ VPC is the foundation of your network in AWS.
- ✓ Route 53 helps users find your resources.
- ✓ ELB ensures high availability and scalability.
- ✓ CloudFront brings your content closer to users.
- ✓ Direct Connect & VPN connect your on-premise network to AWS securely.



TIP

Design your network with security, scalability and performance in mind. Always follow the Well-Architected Framework!



QUICK RECAP

- VPC → Isolated Network
- Route 53 → DNS Routing
- ELB → Traffic Distribution
- CloudFront → CDN
- Direct Connect → Private Link
- VPN → Secure Connection



LIKE



SHARE



COMMENT



SAVE

@ abhi_techhub



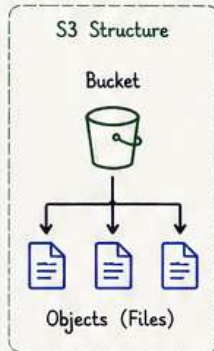
15. AWS STORAGE SERVICES

AWS provides a range of storage services to store, secure and retrieve your data in the cloud.



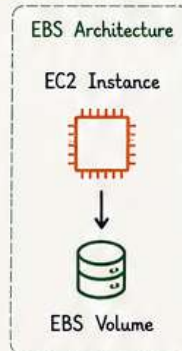
1. AMAZON S3 (SIMPLE STORAGE SERVICE)

- Object storage service.
- Stores data in buckets as objects.
- Highly durable, scalable and available.
- Use cases: Backups, static websites, data lakes.



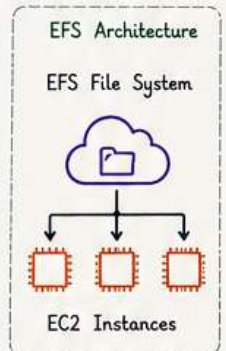
2. AMAZON EBS (ELASTIC BLOCK STORE)

- Block storage for EC2 instances.
- Persistent storage.
- Use cases: Databases, boot volumes, enterprise apps.



3. AMAZON EFS (ELASTIC FILE SYSTEM)

- Managed file storage for Linux.
- Shared across multiple EC2 instances.
- Use cases: Content management, web serving.



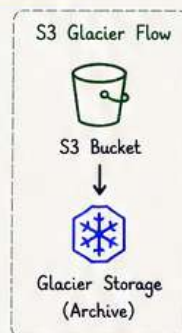
4. AMAZON FSx

- Managed file systems for Windows / Lustre / NetApp / OpenZFS.
- High performance and fully managed.
- Use cases: HPC, media workflows, enterprise apps.



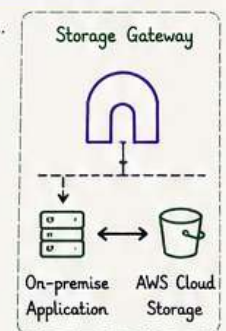
5. AMAZON S3 GLACIER

- Low-cost archival storage.
- For data that is infrequently accessed.
- Options: Instant Retrieval, Flexible Retrieval, Deep Archive.



6. AMAZON STORAGE GATEWAY

- Hybrid storage service.
- Connects on-premise apps to cloud storage.
- Types: File Gateway, Volume Gateway, Tape Gateway.



7. STORAGE SERVICES COMPARISON

Service	Type	Access	Durability	Best For
S3	Object	Internet	99.99999999% (11 9's)	Backups, Data lakes, Static websites
EBS	Block	EC2 Instance	99.999% (4 9's)	Databases, Boot volumes, Enterprise apps
EFS	File (NFS)	Multiple EC2	99.99999999% (11 9's)	Content sharing, Web applications
FSx	File	VPC / On-prem	Varies by service	HPC, Windows apps, Media workflows
S3 Glacier	Object (Archive)	Internet	99.99999999% (11 9's)	Long-term archival storage
Storage Gateway	Hybrid	On-prem / Cloud	Varies by setup	Hybrid cloud storage integration

8. KEY FEATURES SUMMARY

- S3** → Highly scalable object storage.
- EBS** → Block storage for EC2.
- EFS** → Shared file storage.
- FSx** → Fully managed high-performance file systems.
- S3 Glacier** → Low-cost archival storage.
- Storage Gateway** → Bridge between on-prem and cloud storage.

9. STORAGE CLASS IN S3

	Standard	Frequent access, low latency.
	Standard - IA	Infrequent access, lower cost.
	One Zone - IA	Infrequent access, stored in a single AZ.
	Glacier Instant	Archive data, instant retrieval.
	Glacier Flexible	Archive data, minutes to hours retrieval.
	Glacier Deep Archive	Lowest cost, retrieval in hours.

10. BEST PRACTICES

- ✓ Choose the right storage service for your use case.
- ✓ Enable versioning and lifecycle policies in S3.
- ✓ Encrypt data at rest and in transit.
- ✓ Use MFA delete for critical buckets.
- ✓ Monitor storage usage and costs.
- ✓ Regularly review access permissions.



TIP

Use S3 for scale, EBS for performance, EFS/FSx for sharing, Glacier for saving!



COMMON USE CASES

- S3 → Backups, Data lakes, Static websites
- EBS → Databases, Boot volumes
- EFS/FSx → File sharing, HPC
- Glacier → Archival, Compliance



LIKE



SHARE



COMMENT



SAVE



@ abhi_techhub

